Life Support & Atmospherics: System Design Description

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Revision	B
Status	APPROVED
Classification	INTERNAL
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1. System overview

Life support keeps the atmosphere, water, gravity and temperature within crew limits in every inhabited zone.

The system is led by Life Support & Habitation. It is delivered through 6 work packages with a combined budget of 97,195.07 Mcr.

2. Components

Code	Component	Specification	Drawing	Bill of materials
LS-ATP	Atmosphere processor	SHV-LS-SPEC-0101	SHV-LS-DWG-0201	SHV-LS-BOM-0301
LS-WRC	Water reclamation plant	SHV-LS-SPEC-0005	SHV-LS-DWG-0202	SHV-LS-BOM-0302
LS-GRV	Artificial gravity generator	SHV-LS-SPEC-0103	SHV-LS-DWG-0203	SHV-LS-BOM-0303
LS-TRG	Thermal regulation loop	SHV-LS-SPEC-0104	SHV-LS-DWG-0204	SHV-LS-BOM-0304

3. Architecture

3.1 Atmosphere processor

The atmosphere processor is specified in SHV-LS-SPEC-0101. Its governing parameter is the processing rate, set at 8,694 m³/s. The responsible engineer is Wilde Tregar.

3.2 Water reclamation plant

The water reclamation plant is specified in SHV-LS-SPEC-0005. Its governing parameter is the reclamation rate, set at 98.2 %. The responsible engineer is Quinn Calloway.

3.3 Artificial gravity generator

The artificial gravity generator is specified in SHV-LS-SPEC-0103. Its governing parameter is the field uniformity, set at 97.8 %. The responsible engineer is Wren Ambrose.

3.4 Thermal regulation loop

The thermal regulation loop is specified in SHV-LS-SPEC-0104. Its governing parameter is the setpoint, set at 21.8 °C. The responsible engineer is Tyce Easton.

4. Interfaces

Document	Title
SHV-ICD-PD-LS	Interface Control Document: Power Distribution / Life Support & Atmospherics

Document	Title
SHV-ICD-LS-HA	Interface Control Document: Life Support & Atmospherics / Habitation Blocks
SHV-ICD-LS-WM	Interface Control Document: Life Support & Atmospherics / Waste Management & Recycling
SHV-ICD-LS-AG	Interface Control Document: Life Support & Atmospherics / Hydroponics & Food Production

5. Work packages

Work package	Name	Owner	Status	Complete
WP-SHV-LS-01	Life Support & Atmospherics: Design & analysis	Theron Wolcott	COMPLETE	100.0%
WP-SHV-LS-02	Life Support & Atmospherics: Fabricate & install atmosphere processor	Ines Everhart	IN_PROGRESS	96.8%
WP-SHV-LS-03	Life Support & Atmospherics: Fabricate & install water reclamation plant	Noa Venner	IN_PROGRESS	83.9%
WP-SHV-LS-04	Life Support & Atmospherics: Fabricate & install artificial gravity generator	Treva Bresk	IN_PROGRESS	84.1%
WP-SHV-LS-05	Life Support & Atmospherics: Fabricate & install thermal regulation loop	Nerys Chalke	IN_PROGRESS	63.8%
WP-SHV-LS-06	Life Support & Atmospherics: Integration & test	Sola Quarrel	IN_PROGRESS	66.7%

6. Risks

- Supply of atmosphere processor assemblies depends on a single supplier.
- Supply of water reclamation plant assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.